② a) Matriks a + Determinan

Matriks A

$$A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix}$$

Substitusi $a = 0$, $b = 3$

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 3 \end{bmatrix}$$

Det (ekspansi baris pertama)

$$\det(A) = 1 \begin{vmatrix} 2 & 3 \\ 1 & 3 \end{vmatrix} - 3 \begin{vmatrix} 1 & 3 \\ 3 & 3 \end{vmatrix} + 2 \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix}$$

Hit : $1(2.3 - 3.1)$
$-3(1.3 - 3.3)$
$+ 2(1.1 - 2.3)$

$= 1(6-3) - 3(3-9) + 2(1-6)$
$= 3 - 3(-6) + 2(-5)$
$= 3 + 18 - 10 = 11$

$\det(A) = 11$

b) Minor (wajib tabel Dikertas)

Hit semua : $M_{11} = \begin{vmatrix} 2 & 3 \\ 1 & 3 \end{vmatrix} = 3$

$M_{12} = \begin{vmatrix} 1 & 3 \\ 3 & 3 \end{vmatrix} = -6$

$M_{13} = \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} = -5$

Baris 2 : $M_{21} = \begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix} = 7$

$M_{22} = \begin{vmatrix} 1 & 2 \\ 3 & 3 \end{vmatrix} = -3$

$M_{23} = \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} = -8$

Baris 3 : $M_{31} = \begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} = 5$

$M_{32} = \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} = 1$

$M_{33} = \begin{vmatrix} 1 & 3 \\ 1 & 2 \end{vmatrix} = -1$

c) Kofaktor

Gunakan : $C_{ij} = (-1)^{i+j} M_{ij}$

Hasil : $C = \begin{vmatrix} 3 & 6 & -5 \\ -7 & -3 & 8 \\ 5 & -1 & -1 \end{vmatrix}$

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d) Adjoin

Transpose :

$$Adj(A) = \begin{bmatrix} 3 & -7 & 5 \\ 6 & -3 & -1 \\ -5 & 8 & -1 \end{bmatrix}$$

e) Invers :

$$A^{-1} = \frac{1}{11} \begin{bmatrix} 3 & -7 & 5 \\ 6 & -3 & -1 \\ -5 & 8 & -1 \end{bmatrix}$$

F) Studi kasus P

$$P = \begin{bmatrix} 120 & 53 & 200 \\ 86 & 2 & 13 \\ 33 & 90 & 170 \end{bmatrix}$$

ekripsi

$$E = A.P \rightarrow P' = A^{-1}.E$$

Hasil : $P' = P$

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